1. Online convex optimization

1. Protocol and fixed comparator

Let $\mathcal{K} \subseteq \mathbb{R}^d$ be nonempty, closed, and convex, and let the allowed losses be bounded convex functions $f_t : \mathcal{K} \to \mathbb{R}$. On round t:

1. choose $x_t \in \mathcal{K}$ before seeing f_t;
2. reveal f_t and incur $f_t(x_t)$.

For the realized sequence,

$$R_T = \sum_{t=1}^{T} f_t(x_t) - \inf_{x\in\mathcal{K}} \sum_{t=1}^{T} f_t(x).$$

The comparator is one point fixed across all rounds and is evaluated on the same revealed losses. Write a minimum or "best fixed point" only when the infimum is attained. An adaptive loss chosen after x_t is permitted by the declared protocol; choosing x_t after seeing f_t is not.

2. Convex loss linearization

If f_t is differentiable and convex, then for every feasible comparator x,

$$f_t(x_t) - f_t(x) \le \nabla f_t(x_t)^T (x_t - x).$$

This converts a nonlinear loss gap into a gradient inner product. It is an upper bound licensed by differentiable convexity; it does not say that the current gradient points toward the hindsight comparator.

3. Euclidean projection

For nonempty closed convex $\mathcal{K}$,

$$\Pi_{\mathcal{K}}(y) = \arg\min_{x\in\mathcal{K}} \|x - y\|_2.$$

The closest point is unique, and for every feasible z,

$$\|\Pi_{\mathcal{K}}(y) - z\|_2 \le \|y - z\|_2.$$

Feasibility of z and convexity of the set are essential. Projection makes an outside proposal feasible, but the theorem does not give a universal running time: cost depends on how $\mathcal{K}$ is represented.

2. Projected online gradient descent

1. Update and one-step inequality

After observing the current differentiable convex loss, set $g_t = \nabla f_t(x_t)$ and update

$$x_{t+1} = \Pi_{\mathcal{K}}(x_t - \eta_t g_t), \quad \eta_t > 0.$$

For a constant $\eta > 0$ and any $x \in \mathcal{K}$, projection and squared-distance expansion give

$$g_t^T(x_t - x) \le \frac{\|x_t - x\|_2^2 - \|x_{t+1} - x\|_2^2}{2\eta} + \frac{\eta}{2}\|g_t\|_2^2.$$

Combine this with convex linearization before summing: regret contains loss gaps, while squared distance directly controls only gradient inner products.

2. Telescoping and the OGD theorem

Summing the one-step bound leaves only endpoints:

$$\sum_{t=1}^{T}(f_t(x_t) - f_t(x)) \le \frac{\|x_1 - x\|_2^2 - \|x_{T+1} - x\|_2^2}{2\eta} + \frac{\eta}{2}\sum_{t=1}^{T}\|g_t\|_2^2.$$

If $T \ge 1$, $\mathcal{K}$ has diameter at most D, and $\|\nabla f_t(z)\|_2 \le G$ for every t and feasible z, then projected OGD satisfies, for every $\eta > 0$,

$$R_T \le \frac{D^2}{2\eta} + \frac{\eta G^2 T}{2}.$$

When $D > 0$ and $G > 0$, choose $\eta = \frac{D}{G\sqrt{T}}$ to obtain $R_T \le DG\sqrt{T}$.

3. Parameter and computation gates

Case	Correct conclusion
$D = 0$	$\mathcal{K}$ is a singleton, hence $R_T = 0$; do not use $\eta = 0$.
$G = 0$	Each differentiable loss is constant on convex $\mathcal{K}$, hence $R_T = 0$; do not divide by zero.
$D, G > 0$	The balanced positive step is legal.

No explicit d in $DG\sqrt{T}$ does not mean dimension-free: D and G may scale with d. The regret proof also does not determine projection cost.

3. From unstable leaders to FTRL

1. FTL and its instability

Follow the Leader chooses only from past revealed losses:

$$x_t \in \arg\min_{x\in\mathcal{K}} \sum_{s=1}^{t-1} f_s(x).$$

A tie-breaking rule is part of the implementation. Including f_t in this objective before playing x_t violates the online information order.

FTL can have linear regret. On $\mathcal{K} = [-1, 1]$, take $f_1(x) = \frac{x}{2}$ and, for $t \ge 2$, alternate $-x$ on even rounds and x on odd rounds. With the stated round-1 tie choice $x_1 = 1$, the past leader alternates endpoints and has exact regret T. Changing only the first tie cannot remove the later jumps.

2. Strong convexity is curvature

Relative to a declared norm $\|\cdot\|$, differentiable $\mathcal{R}$ is α-strongly convex, $\alpha > 0$, when

$$\mathcal{R}(y) \ge \mathcal{R}(x) + \nabla\mathcal{R}(x)^T(y - x) + \frac{\alpha}{2}\|y - x\|^2.$$

For nonsmooth convex $\mathcal{R}$, use a valid subgradient at x. Strong convexity supplies a quadratic lower gap; it is not smoothness, differentiability, or uniqueness of a gradient. On $[-1, 1]$, $\mathcal{R}(x) = \frac{x^2}{2}$ is 1-strongly convex in the Euclidean norm, whereas $|x|$ is convex and nonsmooth but not strongly convex.

For a twice-differentiable function on a convex Euclidean domain, Hazan's regularizer assumptions can be written

$$\alpha I \preccurlyeq \nabla^2\mathcal{R}(x) \preccurlyeq \beta I \quad \text{for every} \quad x.$$

The lower Hessian bound certifies strong convexity; the upper bound certifies gradient-Lipschitz β-smoothness. Neither property implies the other, and the Hessian test does not replace the subgradient definition for nonsmooth strongly convex functions.

3. Linearized FTRL update

Given $\eta > 0$ and a strongly convex regularizer $\mathcal{R}$,

$$x_t = \arg\min_{x\in\mathcal{K}} \left[\eta \sum_{s=1}^{t-1} g_s^T x + \mathcal{R}(x)\right], \quad g_s = \nabla f_s(x_s).$$

Only past linearized gradients may enter x_t. The regularizer makes the leader resist small changes in cumulative loss. In the signed example with $\mathcal{R}(x) = \frac{x^2}{2}$ and $0 < \eta \le 2$, $x_t = \mathrm{clip}\left(-\eta \sum_{s<t} g_s, -1, 1\right)$: later decisions alternate between $-\frac{\eta}{2}$ and $\frac{\eta}{2}$, rather than between endpoints. Stability alone does not license logarithmic regret.

4. FTRL regret tradeoff

Let x_1 minimize $\mathcal{R}$ and $\Delta_u = \mathcal{R}(u) - \mathcal{R}(x_1) \ge 0$. Under the source's bounded closed convex set and twice-differentiable, α-strongly-convex, gradient-Lipschitz β-smooth regularizer setup, the registered FTRL update obeys

$$R_T(u) \le 2\eta \sum_{t=1}^{T} \left(\|g_t\|_t^*\right)^2 + \frac{\Delta_u}{\eta}.$$

Here $\|\cdot\|_t$ is the local norm induced by the regularizer curvature and $\|\cdot\|_t^*$ is its dual; these are not automatically the global Euclidean norm. The two terms are respectively the movement bill and comparator regularizer geometry.

If $\|g_t\|_t^* \le G_{\mathcal{R}}$ and $\Delta_u \le \Delta_{\mathcal{R}}$ uniformly, then

$$R_T(u) \le 2\eta G_{\mathcal{R}}^2 T + \frac{\Delta_{\mathcal{R}}}{\eta}.$$

For positive $\Delta_{\mathcal{R}}, G_{\mathcal{R}}$, balancing with $\eta = \sqrt{\frac{\Delta_{\mathcal{R}}}{2G_{\mathcal{R}}^2 T}}$ gives order $\sqrt{T}$.

4. Which rates are licensed?

1. Match structure, schedule, and method

Loss structure	Licensed mechanism	Rate
general convex	fixed strong $\mathcal{R}$ + balanced FTRL	$O\left(\sqrt{T}\right)$
α-strongly convex	projected OGD, $\eta_t = \frac{1}{\alpha t}$	$R_T \le \frac{G^2}{2\alpha}(1+\log T)$
α-exp-concave	$\exp(-\alpha f_t)$ concave + matching EWOO/ second-order method	$O(\log T)$

For strongly convex losses, the negative term $-\alpha\frac{\|x_t-u\|_2^2}{2}$ cancels successive distance coefficients under the matching schedule, leaving $\sum_t \frac{1}{t} \le 1 + \log T$. A strong regularizer alone, a decreasing step without loss curvature, or exp-concavity paired with an arbitrary first-order update does not prove a logarithmic rate.

2. General square-root lower bound

For every OCO algorithm there exists a bounded convex decision problem with bounded linear gradients whose worst-case regret is

$$\Omega\left(DG\sqrt{T}\right).$$

One stationary construction uses $\mathcal{K} = \{x \in \mathbb{R}^n : \|x\|_\infty \le 1\}$ and independent uniform sign vectors $v_t \in \{-1,1\}^n$ with $f_t(x) = v_t^T x$. The learner's expected loss is zero because v_t is fresh after x_t is fixed. The hindsight comparator loses $-\sum_i \left|\sum_t v_t(i)\right|$, whose expectation has order $-n\sqrt{T}$. Since $D \le 2\sqrt{n}$ and $G \le \sqrt{n}$, this matches the $DG\sqrt{T}$ scale up to constants.

The result applies to unrestricted bounded OCO; it does not contradict faster rates under additional curvature or exp-concavity assumptions.

5. From online regret to population risk

1. A valid online-to-batch bridge

Sequence-wise regret and population risk are different objects. A valid bridge assumes:

1. iid examples z_t from one distribution;
2. a convex decision/hypothesis class and bounded loss $\ell \in [0,1]$ convex in the decision;
3. h_t is chosen before z_t, and $\bar{h} = T^{-1}\sum_t h_t$;
4. concentration for the adaptive iterates and fixed comparator.

For an arbitrary fixed comparator h, define

$$\mathrm{Regret}_T(h) = \sum_{t=1}^{T} \ell(h_t, z_t) - \sum_{t=1}^{T} \ell(h, z_t).$$

Under the source's reduction, with probability at least $1-\delta$,

$$R\left(\bar{h}\right) \le R(h) + \frac{\mathrm{Regret}_T(h)}{T} + \sqrt{\frac{8\log\left(\frac{2}{\delta}\right)}{T}}.$$

Convexity licenses averaging; iid sampling and boundedness support the risk and concentration steps. Sublinear regret alone is not a population generalization theorem, and $\frac{\mathrm{Regret}_T(h)}{T}$ is not the same object as $R(h)$.

2. Keep algorithm identities separate

Method	Update object	Do not identify with
OGD	gradient step + Euclidean projection	general FTRL without matching geometry
FTRL	past linearized gradients + $\mathcal{R}$	FTL on original nonlinear losses
Hedge	exponential expert weights	linear MWU by name alone
linear MWU	$w_{t+1}(i) = w_t(i)(1-\eta\ell_t(i))$	Hedge or AdaBoost by name alone
AdaBoost	training-example distribution + weak learner	an OCO method with identical iterates

Shared potential reasoning or comparator language does not make update states, losses, or iterates identical.